04 | Full attention = Query routing + Key routing + Value

Target o1 in the same two-token chain; head dimension = 1, bias = 0; four input coordinates.

$$dO = J_{softmax}(S)[(dQ)K^T + Q(dK)^T]V + A(dV)$$

Endpoint components

Query	Key	Value	Full attention	Fixed-A endpoint
0.210 0.210	-0.105 -0.105	0.060 0.060	0.165 0.165	0.060 0.060
0.000 0.000	0.315 0.315	1.321 1.321	1.636 1.636	1.321 1.321
$\Sigma = 0.4200$	$\Sigma = 0.4200$	$\Sigma = 2.7616$	$\Sigma = 3.6015$	$\Sigma = 2.7616$

Path-integrated components

Path Query	Path Key	Path Value	Path Full attention	Fixed-A Path IG
0.117 0.117	-0.058 -0.058	0.176 0.176	0.235 0.235	0.060 0.060
0.000 0.000	0.175 0.175	0.971 0.971	1.146 1.146	1.321 1.321
$\Sigma = 0.2332$	$\Sigma = 0.2332$	$\Sigma = 2.2951$	$\Sigma = 2.7616$	$\Sigma = 2.7616$

Each of the first two rows closes: Q + K + V = full. The right column instead freezes clean A.

Here V is linear and the baseline is zero, so fixed-A Endpoint = fixed-A IG = the endpoint Value component.

That last equality is specific to this example. In general, conditional Value-path IG differs from full-path Value conductance.